

```

...amples/hailo-apps-infra-main/hailo_apps_infra -- ssh bera@10.195.80.97
[venv_hailo_rpi5_examples] bera@raspberrypi:~/hailo-rpi5-examples/hailo-apps-infra-main/hailo_apps_infra $ python lpr3_pi5.py sample.png
Using GPU. Note: This module is much faster with a GPU.

0: 224x320 4 license_plates, 155.9ms
Speed: 4.9ms preprocess, 155.9ms inference, 1.2ms postprocess per image at shape (1, 3, 224, 320)
4 License Plate(s) Found:
0: 448x640 4 cars, 1 truck, 468.5ms
Speed: 7.0ms preprocess, 468.5ms inference, 4.9ms postprocess per image at shape (1, 3, 448, 640)
Detected Plate: Z (24.6%)
Detected Plate: BN6255X (78.6%)
Detected Plate: SP1044K (35.9%)
Detected Plate: KRE (44.8%)
[venv_hailo_rpi5_examples] bera@raspberrypi:~/hailo-rpi5-examples/hailo-apps-infra-main/hailo_apps_infra $

```

Fig. 2: Terminal output



Fig. 3: Sample.png, AI-generated image used for testing

overclocking for improved performance. The resulting optimised configuration ensures consistent, efficient operation while meeting the real-time requirements of object detection and number-plate recognition.

The required components for building the system are listed in Bill of Materials table.

Software requirements

1. Operating system. Raspberry Pi OS (64-bit recommended for better performance)

2. Python. Version 3.11 or higher

3. AI models. kbest.pt and yolov8n.pt

4. Tesseract-OCR. Required for number plate text extraction

5. Virtual environment. Recommended for managing Python dependencies and avoiding conflicts

6. Python libraries. OpenCV, Ultralytics YOLO, NumPy, and Pillow

A detailed requirements.txt file is provided. After creating and activating the virtual environment, the dependencies are installed using:

```
pip install -r requirements.txt
```

If the HailoRT AI accelerator is also used on the same system, the provided requirements.txt file should be installed without modification. It has been configured to prevent dependency conflicts, particularly

with the NumPy version. Installing a higher NumPy version may cause compatibility issues.

Configuration

The Raspberry Pi must be overclocked to handle the processing load required for real-time YOLO-based object detection. Overclocking improves performance but also increases heat generation and potential long-term hardware stress. Proper cooling and temperature monitoring are therefore essential. Follow the steps below for configuration:

1. Creating a temperature monitoring script.

```

$ nano temp
clear
sensors
vcgencmd measure_temp
vcgencmd measure_clock arm

```

Create an overclock control script:

```

$> nano overclock
sudo cpufreq-set -g performance
sudo cpufreq-set -g ondemand
sudo cpufreq-set -g powersave
vcgencmd measure_clock arm
vcgencmd measure_temp

```

2. Enabling overclocking.

```

sudo nano /boot/firmware/config.txt
$> sudo nano /boot/firmware/config.txt
[reboot after each changes] [overclock]
over_voltage=4
arm_freq=2500

```

```
gpu_freq=750
```

Reboot the system after each change.

The on-demand performance mode allows the CPU clock frequency to vary between 1.5GHz and 2.5GHz, while the fan RPM is automatically regulated through thermal management.

3. Installing the required monitoring tools.

Install the lm-sensors package:

```

sudo apt-get install lm-sensors
$>sensors # to show cooler fan rpm & CPU
temperature.

```

Check CPU temperature and fan RPM:

```

sensors
Make both scripts executable:

```

```
chmod +x temp overclock
```

Run them using:

```

$> ./overclock
$> ./temp

```

4. Virtual environment setup.

After configuring overclocking, set up a Python virtual environment. This ensures that all dependencies remain isolated from the system Python installation. If anything breaks, you can simply remove the environment without affecting the OS.

By default, Python 3.11 is installed on Raspberry Pi 5. If not, install it:

```

$>sudo apt install -y python3.11
python3.11-venv python3-pip

```

Create and activate a virtual environment:

```

python3.11 -m venv myenv
source myenv/bin/activate

```

Install the required packages:

```
pip install -r requirements.txt
```

5. Camera and Wi-Fi setup on Raspberry Pi 5.

Ensure the camera is enabled:

```
$>sudo raspi-config
```

Go to Interface Options

- Enable camera
- Reboot, if required

If the camera option is not visible, install libcamera:

```
$>sudo apt install libcamera-apps
```

Verify installation:

```
$>libcamera-jpeg -version
```

6. For Wi-Fi setup.

```
$>sudo raspi-config
```